INTRO TO STATISTICS

It's all about interpreting data (Table, list, graphs, etc)

Data can be form

Population (the entire list of a specified group)

Mean (a subset of the Population)

We usually investigate a small sample of the population to draw conclusions for the whole population itself.

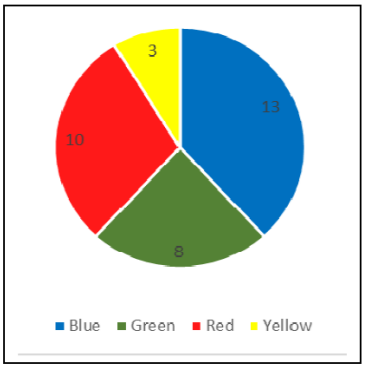
Discrete	OR	Continuous
$\{10, 20, 30\}$		$[40, 100]$
$\{0, 1, 2, 3, \ldots\}$		$\mathbb{R}$
(finite or numerable set)		(interval)

Data can be organized in several ways. We present some examples below

Frequency table

Colored Balls	Frequency
Blue	13
Green	8
Red	10
Yellow	3

Pie chart



Colored Balls	Freq
Blue	13
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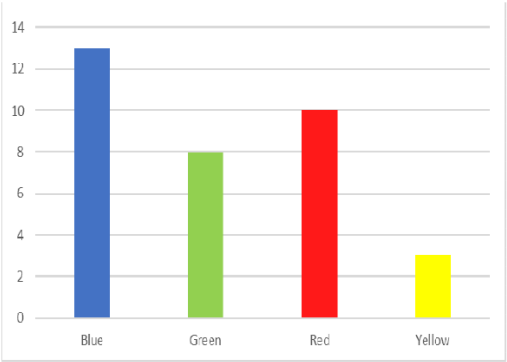
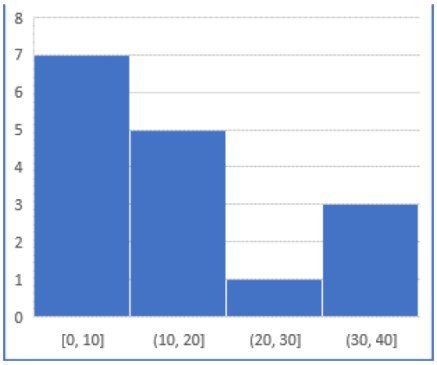
Red	10
Yellow	3

Bar graph (for discrete data)



Age	Frequency
[0,10)	7
[10,20)	5
[20,30)	1
[30,40)	3

Histogram (for continuous data)



Stem and leaf Diagram

Data
12, 14, 16, 16, 20, 21
21, 21, 25, 32, 39, 40
43, 44, 47, 48, 49, 53

Key: 1|3 represents 13

Stem	Leaf
1	2, 4, 6, 6
2	0, 1, 1, 1, 5
3	2, 9
4	0, 3, 4, 7, 8, 9
5	3

Ex: Age of Students in class:

16,15,17,15,15,16,18,15

Mean:

Notation:

Example: The data below is listed in ascending order:

5, 6, 7, 7, 9, 9, 10, 10, 13, 13, 13, 13, 18

a. What is the mean?

Median:

b. What is the median of the following data above?

Mode:

Percentile:

Quartile:

Interquartile range (IQR):

Ex: A data set has the following:

25th percentile = 7

50th percentile = 9.5

75th percentile = 13

Mean (Formally):

Example: The height of dinosaurs is shown below:

Height h(m)	$0 < h \leq 10$	$10 < h \leq 14$	$14 < h \leq 18$	$18 < h \leq 22$
Frequency	5	2	1	2

What is the mean dinosaur height?

SAMPLING AND OUTLIERS

Discrete data:

Continuous data:

Sample:

Reliable data:

Sampling techniques:

A) Simple Randon — Equal chance of choosing out of a hat, number generator, etc.
Ex. Poll students from school-# assigned to Students, choose with random # generator.

B) Convenience — Choose easiest people to sample: ask your friends.
Problems? May not be representative of population.

C) Systematic	Choose random starting point, use fixed interval. Ex. Make a list of all students in a class, choose every 3rd student…
D) Quota	Sample sizes in proportion to who you're polling Ex. 55% girls, 45% boys in school, so sample should have those same %.
E) Stratified	Spilt into strata (smaller groups) Ex. Choose half DP1, half DP2 students